Quiet Neurons: a BDH's layer 2 quietens on what it just learned, not on what is merely predictable

DataForge 2026, Pathway track · concept **Sparse Non-Negative Activations** · Nilay Toshniwal, solo · artifact **nilaymastaadmi.github.io/quiet-neurons** · code **github.com/nilaymastaadmi/quiet-neurons**

THE DESIGN PRESSURE

A dense Transformer evaluates every feed-forward unit for every token; activation sparsity is the counter-proposal. But Transformer sparsity is already emergent [3], needs a hard rectifier for exact zeros [4], and is input-dependent in *which* units fire, not merely how many [5]. Sparsity alone distinguishes nothing; what it depends on does.

WHAT BDH CHANGES

Dragon Hatchling (BDH) [1] makes it structural. In BDH-GPU, which Pathway open-sources, every layer multiplies two rectified (ReLU) vectors elementwise, and that product can never be denser than either. *Active* means a strictly non-zero entry there; a rectifier emits no negatives. That product is the paper's $y_{t,l}$: Eq. (8) defines it as $(D_y LN(\rho x))_+ \odot x$, Appendix E reads `relu(self.ln(a) @ decoder_y) * x`, and Fig. 14 counts its non-zeros. Naming collision: the code's `y_sparse` is the paper's rectified *intermediate*; the paper's y is the code's `xy_sparse`. Attention is the second change: query and key are one tensor, so the score matrix is a causally-masked Gram matrix of activations with **no softmax**.

THE CLAIM, AND THE CORRECTION IT NEEDED

Section 6.4 reports "fewer neurons are active ... for more predictable input signals": at layer 2 of a 65,536-neuron model, 4.0–7.5% active while memorising a new word against about 2.5% while repeating it, a 1.6–3.0× ratio [1, Fig. 14]. We reproduced that protocol on three models trained here from the public code, all for 2,309 steps; it survives a 32-fold shrink: **1.47×, 2.12× and 1.87×** at 2,048, 8,192 and 16,384 neurons, each 1,280 sequences over five independently seeded samples, ±0.005 half-width. **On the paper's own protocol our three models reproduce its result.** This model goes quiet on words it has just picked up from the text in front of it; it does not go quiet just because text is easy to guess. From the outside those two look identical, and they are not the same thing. That they come apart is what we show; why, we do not.

The same measurement narrows what that can mean. Two blocks of one sequence are predicted almost equally well yet differ 2.65-fold in activity, over 1,280 sequences of the 8,192-neuron model: the 13-letter warm-up runs at **9.65%** of layer 2 active at 0.0004 nats of mean surprise, the 56 repeated letters at **3.64%** at 0.004 nats. The warm-up is identical in every training sequence, so it lives in the **weights**; the repeated word is new each time, so it can only come from **context**. What *covaries* with the difference is not predictability but **where the knowledge came from** — a correlation, not an established cause. A dense Transformer trained identically, to a slightly better loss, shows none of it: all eight ratios 0.83 to 1.02 against layer 2's 2.35.

System	Where its sparsity comes from	Carried per token	Evidence status
Dense Transformer feed-forward	Emergent in training [3]; needs a hard rectifier for exact zeros [4]	A key-value cache, growing with context	Extensive, independent; trained here, no signature (0.83–1.02)
Linear attention / SSM (Mamba [6])	None by design; dense recurrent state updates	A fixed-size recurrent state	Extensive, independent
BDH-GPU, measured here	Structural: elementwise product of two rectified vectors, never denser than either	A fixed-size synaptic state	Reproduced here at toy scale; author-reported at 65,536 neurons
BDH-CQ	Not public	Not public	Developer-reported only

EVIDENCE, LABELLED

Measured here: the three ratios, the counterexample, the surprise-injection check, browser-against-PyTorch parity (6.1e-5 max logit difference, sparsity series exact) — an implementation check, not architectural evidence. **Developer-reported, not reproduced:** the Fig. 14 band; 97.4% on Sudoku Extreme, which Pathway says does not reproduce from the open code; BDH-CQ's ARC-AGI-1 result [2]. **Not modelled:** BDH-CQ's internals, "proprietary" by its authors [2]; **it has no role here**. **Where it is likely worse:** sparse non-negative activations plausibly cost capacity against dense superposition; no published comparison runs past GPT-2 scale.

LIMITATIONS AND MATURITY

Depth matters: layer 0 runs *backwards* at all three sizes, ratios below 1 (0.80, 0.80, 0.86): *more* neurons fire while repeating; layer 1 is positive at every size (1.22, 1.41, 1.33) and layer 3 is flat below 16,384. Scaling is **not monotonic**; we published the opposite before the third model finished. The wall-clock confound is gone, the short two retrained to 2,309 steps, the dip 0.2459 at **40×** the mean of half-widths 0.0081 and 0.0042; none of the three published models finished the 4,000-step schedule, though separate runs that did finish move layer 2 only 1.4705 to 1.4620 at n=2,048, but 2.1201 to 2.2000 at n=8,192, a rise of 0.0799 against that run's own ±0.0086 spread. These are toys, 0.4M to 6.3M parameters on a synthetic task. Surprise and sparsity do *not* track per letter (Pearson 0.30 across phases, Spearman −0.05 letter by letter), killing a stronger claim. Most importantly: signature, not cause. Four conditions vary how much of the word sits in the weights. Ranked by first-sight surprise, they **step** rather than rise: mem/rep 1.00 and 1.01 when the word is already learnt, 1.55 and 1.47 when it must be read. K=16 shows nothing because 89% of its word is in the weights; the fixed-word control is degenerate. The 0.73-nat condition is **4.5× easier by loss (0.0384 against 0.1739) yet shows the larger effect**, so difficulty is not the driver. We predicted a rise and got a step. **Maturity:** credible and outsider-checkable for this property; the wider programme is earlier, its headline results developer-reported.

[1] Kosowski, Uznański, Chorowski, Stamirowska, Bartoszkiewicz, *The Dragon Hatchling*, arXiv:2509.26507 (2025), §6.4 and Fig. 14. [2] Engdahl, Kosowski, Chorowski et al., *BDH-CQ*, arXiv:2608.09888 (2026). [3] Li, You, Bhojanapalli et al., *The Lazy Neuron Phenomenon*, arXiv:2210.06313 (2022), ICLR 2023. [4] Mirzadeh, Alizadeh, Mehta et al., *ReLU Strikes Back*, arXiv:2310.04564 (2023). [5] Liu, Wang, Dao et al., *Deja Vu*, arXiv:2310.17157 (2023), ICML 2023. [6] Gu, Dao, *Mamba*, arXiv:2312.00752 (2023).

Where to continue, in order. [1] §6.4 and Fig. 14 for the result reproduced here, then [1] §3.2 and Eq. (8) for the equations defining what is counted, then [3] for why sparsity alone distinguishes no architecture, then the artifact for the counterexample, which no paper states. — Reproduce the n=2,048 row: **python measure.py --ckpt checkpoints/bdh_n2048.pt --embd 64 --mult 32 --repeats 5**.